BIGKAS-AI Offline Rehearsal Runbook

Laptop as the server · HTTPS on the LAN · local Whisper & classifier · prepared 2026-10-22

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Facts this runbook depends on. Laptop Wi-Fi IP when set up: `10.38.134.4` (it will differ on the school router — see section 6). Project: `F:\MyApp\BIGKAS-AI`. XAMPP: `F:\xampp`. mkcert: `F:\MyApp\_tools\mkcert`. Backups: `F:\MyApp\_backups`.

1. What we changed on this machine

ITEM	WHAT / WHERE	ROLLBACK (SEE SECTION 8)
Database backup	`F:\MyApp\_backups\bigkas_ai-20261022-1210.sql` — full dump of the fake test data (39 tables) taken *before* the wipe.	R1
Files backup	`storage-app-20261022-1210.tgz` — the fake audio recordings.	R1
Data wipe	Removed all learners, teachers, parents, assessments (+results, sessions, batches), messages, notifications, activity logs, sessions and the audio files. Class teacher assignments cleared. kept admin, school, 5 classes, 19 reading materials, 8 comprehension questions, 20 interventions, 174 practice items, 12 badges, 13 settings.	R1
mkcert	Portable `mkcert.exe` v1.4.4 in `F:\MyApp\_tools\mkcert`. `mkcert -install` created a local certificate authority (CA) and added it to the *Windows trust store*. CA files live in `C:\Users\NEC\AppData\Local\mkcert`.	R5
Server certificate	`F:\MyApp\_tools\mkcert\certs\bigkas.pem` + `bigkas-key.pem`, valid for `localhost`, `127.0.0.1`, `10.38.134.4` until Jan 2029.	R5

Apache HTTPS vhost	New file `F:\xampp\apache\conf\extra\httpd-bigkas-ssl.conf` (serves `public/` on port 443, 420 s timeout) + one added `Include` line in `httpd.conf` before `httpd-ssl.conf` .	R3
.env	`APP_URL` changed from `http://localhost/bigkas-ai` to `https://10.38.134.4` .	R4
Windows Firewall	Inbound rule `BIGKAS HTTPS` : TCP 443, Allow, **Private** profile only.	R2
Network profile	Wi-Fi set from Public to **Private** (needed for the rule above).	R2
ML service	Flask app in `ml-service/` , port 5000, loopback only. Whisper `small` (pre-trained, int8, CPU) from the local cache + the weakness classifier. Runs fully offline.	R6

Not changed: application code, migrations, the VPS, `php.ini` . (`max_execution_time` is still 120 s — see troubleshooting if a long reading times out.)

2. Start the system (every test session)

Do these in order. Use **PowerShell** for the commands. You need **four things running**: MySQL, Apache, the ML service, and the scheduler.

1

Connect the laptop to the test router and plug it in.

Windows → Settings → Power: set sleep to *Never* while plugged in. Then check the IP and the network profile:

```
ipconfig | Select-String "IPv4"
Get-NetConnectionProfile
```

The profile must say **Private**. If it says Public (it often does on a new network), run in an *Administrator* PowerShell:

```
Set-NetConnectionProfile -InterfaceAlias "Wi-Fi"
```

If the IP is not `10.38.134.4`, do section 6 first.

2 **Start MySQL and Apache.**

Open the **XAMPP Control Panel** → click *Start* next to **MySQL** and next to **Apache**. Both turn green. (`php artisan serve` is *not* needed any more — Apache serves the app.)

3 **Start the AI & STT service (Whisper + classifier).**

Open a PowerShell window and leave it open:

```
cd F:\MyApp\BIGKAS-AI\ml-service
$env:HF_HUB_OFFLINE = '1'
.\venv\Scripts\python.exe app.py
```

Wait about 30–45 seconds until you see `Whisper 'small' model loaded` and `Serving Flask app 'app'`. `HF_HUB_OFFLINE=1` stops it trying to reach the internet.

> This runs Flask in debug mode (hard-coded in `app.py`). If you edit a Python file while it runs, it reloads and the model takes ~40 s to load again. Don't touch `ml-service/` files during a session.

4 **Start the scheduler** (needed for batch assessments).

Open a second PowerShell window and leave it open:

```
cd F:\MyApp\BIGKAS-AI
```

```
F:\xampp\php\php.exe artisan schedule:work
```

> **Do not skip this.** Batch recordings are scored by `assessments:rescore`, run every minute by the scheduler. Without `schedule:work` they stay at "queued" forever, with no error. `health:check` also runs from here.

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3. Verify before anyone arrives

☐ ML service health — PowerShell:

```
curl.exe -s http://127.0.0.1:5000/api/health
```

Must show `"status": "ok"`, `"whisper_loaded": true`, `"classifier_loaded": true`.

☐ HTTPS on the laptop — open `https://10.38.134.4/login` in Chrome. Padlock, no warning.

(Command-line check needs `curl.exe -s --ssl-no-revoke -o NUL -w "%{http_code}" https://10.38.134.4/login` → `200`; plain curl fails on mkcert certs because of a Windows revocation quirk, not a real fault.)

☐ Log in as **admin**. The *AI* and *STT* badges show online.

☐ From a phone on the same Wi-Fi, open `https://10.38.134.4/login` (see next section).

☐ Make one test recording from a phone and confirm it transcribes. *Done once on 2026-10-22: worked.*

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4. Connect devices (children, teachers)

> **Decision:** children's and teachers' devices will *not* get the CA installed. They accept the one-time warning instead. The microphone still works because the page is still `https://`.

Without the CA (default)

1. Connect the device to the test Wi-Fi (no internet needed).
2. Open **Chrome** (not Firefox) and go to `https://10.38.134.4`.
3. On the warning page tap **Advanced** → **Proceed to 10.38.134.4 (unsafe)**. iPhone Safari: *Show Details* → *visit this website*.
4. When asked, tap **Allow** for the microphone.

> The warning can return after the browser closes, and the mic permission is asked each time. Print a slip for children: *"Tap Advanced → Proceed. It is safe."*

Optional: install the CA on a device (removes the warning)

The file is `C:\Users\NEC\AppData\Local\mkcert\rootCA.pem`. Send **only** this file — **never** `rootCA-key.pem`. Move it by USB, Bluetooth, or temporarily copy it to `F:\MyApp\BIGKAS-AI\public\rootCA.pem` and open `https://10.38.134.4/rootCA.pem` on the device (delete it from `public\` afterwards).

DEVICE	STEPS
Android	Settings → Security (or Security & privacy) → More security settings → Encryption & credentials → *Install a certificate* → *CA certificate* → Install anyway → pick `rootCA.pem` (rename to `rootCA.crt` if it is not listed). Fully restart Chrome.
iPhone / iPad	Open the file, allow the profile download → Settings → General → VPN & Device Management → install "mkcert" → then Settings → General → About → **Certificate Trust Settings** → turn on full trust.
Windows	Double-click `rootCA.pem` → Install Certificate → place in *Trusted Root Certification Authorities*. Restart the browser.
Chromebook	Chrome → Settings → Privacy and security → Security → Manage certificates → Authorities → Import; tick "Trust this certificate for identifying websites".

5. Dress-rehearsal checklist

☐ Admin: create the **real teachers** and attach them to the 5 classes (the assignments were cleared by the wipe).
☐ Create 5–8 **sample learners** across grades 3–6 and a couple of parent accounts. (They can be wiped again afterwards — see R1.)
☐ Connect **5 or more devices** of different kinds (Android, iPhone, tablet, laptop).
☐ Run a **batch assessment**: issue join codes, have several children read at once.
☐ Check on the batch board that recordings move from *queued* to *scored* (this proves the scheduler is running).
☐ **Time it:** submit ~10 readings close together. Note how long the last one takes. Whisper `small` on CPU is slower than real time; the batch board shows a "about N minutes" estimate.
☐ Open the **results page** as a teacher and as a parent for one child.

☐ Stagger the real sessions in groups of 5–8 children, one grade per slot.
☐ Keep a fallback: record first, let scoring catch up later. Nothing is lost.

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6. If the laptop's IP changes (new router)

The IP is written into the certificate, the Apache vhost, and `.env`. The simplest fix is to give the laptop a **fixed IP / DHCP reservation** on the router. Otherwise, with `NEW_IP` as the new address:

1. Reissue the certificate (the CA stays the same, so devices that already trusted it keep trusting it):

```
cd F:\MyApp\_tools\mkcert\certs
F:\MyApp\_tools\mkcert\mkcert.exe -cert-file bigk
```

2. Edit `F:\xampp\apache\conf\extra\httpd-bigkas-ssl.conf` : change `ServerName 10.38.134.4` to `ServerName NEW_IP` .

3. Edit `F:\MyApp\BIGKAS-AI\.env` : `APP_URL=https://NEW_IP` , then:

```
cd F:\MyApp\BIGKAS-AI
F:\xampp\php\php.exe artisan config:clear
```

4. Restart Apache from the XAMPP Control Panel. Make sure the new Wi-Fi network is **Private** (section 2, step 1).

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7. Shut down at the end of a session

1. Close the scheduler window (`Ctrl+C`).
2. Close the ML service window (`Ctrl+C`). If it was started in the background and the window is gone, stop it by port:

```
Get-NetTCPConnection -LocalPort 5000 -State Listen | ForEach-Object { Stop-Process -Id $_.Owning
```

3. XAMPP Control Panel: *Stop* Apache, then MySQL.
4. Take a backup of the real data collected (do this **after every session**):

```
$ts = Get-Date -Format yyyyMMdd-HHmm
F:\xampp\mysql\bin\mysqldump.exe -u root --single-transaction --routines --triggers bigkas_ai >
tar -czf F:\MyApp\_backups\session-files-$ts.tgz -C F:\MyApp\BIGKAS-AI storage/app
```

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8. Rollback

> **R1 overwrites the current database.** It restores the *fake* data. If you have already collected real data, take a fresh backup first (section 7, step 4) and do not run R1.

R1 — Restore the original fake data (database + audio)

```
# 1. Restore the database (the dump drops and recreates every table)
F:\xampp\mysql\bin\mysql.exe -u root bigkas_ai < F:\MyApp\_backups\bigkas_ai-20261022-1210.sql

# 2. Restore the audio files
tar -xzf F:\MyApp\_backups\storage-app-20261022-1210.tgz -C F:\MyApp\BIGKAS-AI

# 3. Clear caches
cd F:\MyApp\BIGKAS-AI
F:\xampp\php\php.exe artisan optimize:clear
```

Verify: `F:\xampp\mysql\bin\mysql.exe -u root bigkas_ai -e "select count(*) from learners"` should return **44**.

R1b — Wipe the data again (back to a clean slate for the real test)

If you restored fake data and want it empty again, or want to clear the *rehearsal* data before the real sessions, ask Claude to re-run the wipe: it removes learners, non-admin users, assessments and their dependents, and the audio files, and keeps reading materials, interventions, practice items, badges, settings, school, classes and the admin.

R2 — Windows Firewall and network profile

Administrator PowerShell:

```
Remove-NetFirewallRule -DisplayName "BIGKAS HTTPS"
Set-NetConnectionProfile -InterfaceAlias "Wi-Fi" -NetworkCategory Public
```

Verify: `Get-NetFirewallRule -DisplayName "BIGKAS HTTPS"` should say it cannot find the rule.

R3 — Apache HTTPS vhost

```
Copy-Item F:\xampp\apache\conf\httpd.conf.bak-bigkas F:\xampp\apache\conf\httpd.conf -Force
Remove-Item F:\xampp\apache\conf\extra\httpd-bigkas-ssl.conf
F:\xampp\apache\bin\httpd.exe -t
```

It should print `Syntax OK`. Then restart Apache from the XAMPP Control Panel. (The backup was taken right before the change, so this undoes only the vhost line. If you have edited `httpd.conf` since, remove just the line `Include conf/extra/httpd-bigkas-ssl.conf` instead.)

R4 — .env

```
Copy-Item F:\MyApp\_backups\env-20261022.bak F:\MyApp\BIGKAS-AI\.env -Force
cd F:\MyApp\BIGKAS-AI
F:\xampp\php\php.exe artisan config:clear
```

This restores `APP_URL=http://localhost/bigkas-ai`. It also reverts any other `.env` change since 2026-10-22 — if you changed anything else, edit just the `APP_URL` line by hand instead.

R5 — mkcert, the certificate, and the trusted CA

```
# Remove the CA from the Windows trust store (and Firefox/Java stores if present)
F:\MyApp\_tools\mkcert\mkcert.exe -uninstall

# Delete the server certificate and the tool
Remove-Item -Recurse -Force F:\MyApp\_tools\mkcert\certs
# Optional: delete the CA itself, INCLUDING its private key. Do this only when you are finished with H
# Remove-Item -Recurse -Force "$env:LOCALAPPDATA\mkcert"
# Remove-Item -Recurse -Force F:\MyApp\_tools\mkcert
```

Devices where you installed `rootCA.pem`: remove it there too.

- Android: Settings → Security → Encryption & credentials → User credentials → tap the mkcert certificate → Uninstall.
- iPhone/iPad: Settings → General → VPN & Device Management → mkcert → Remove Profile.
- Windows: `Win+R` → `certmgr.msc` → Trusted Root Certification Authorities → Certificates → delete the "mkcert …" entry.

> If you delete the CA's key and later re-run `mkcert -install`, a *new* CA is created and every device that trusted the old one must install the new `rootCA.pem`.

R6 — ML service and scheduler

They change nothing on disk apart from logs in `F:\MyApp\_backups\ml-service.*.log`. Rolling back means stopping them (section 7). The Whisper weights remain in `%USERPROFILE%\.cache\huggingface\hub` and are reused next time.

Full back-out to the original setup

Do R2 → R3 → R4 → R5, then restart Apache, then run the app the old way: `cd F:\MyApp\BIGKAS-AI; php artisan serve`. Add R1 only if you also want the fake data back.

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9. Troubleshooting

SYMPTOM	LIKELY CAUSE	FIX
Phone times out on `https://IP`	Network is Public, or router has client isolation / AP isolation on, or wrong IP.	Set profile to Private (section 2). Check the router's "AP isolation" setting is off. Re-check `ipconfig`.
Admin shows **AI / STT offline**	ML service not running (or still loading).	Start it (section 2, step 3). Wait ~45 s. Check `/api/health`.
Batch recordings stay *queued*	Scheduler not running.	Start `schedule:work` (section 2, step 4).
Browser says the page is *unsafe* / mic does not prompt	Warning not accepted, or an HTTP link was used.	Use `https://`, accept Advanced → Proceed, use Chrome.
Reading fails after ~2 minutes	PHP `max_execution_time` is 120 s; Whisper needs longer.	Raise `max_execution_time` in `F:\xampp\php\php.ini` to 360 and restart Apache. (Not changed yet.)
Apache won't start	Port 443 in use, or config error.	`F:\xampp\apache\bin\httpd.exe -t`; read `F:\xampp\apache\logs\error.log`. Fall back to R3.
Site shows a 500 error	Stale config cache after an `.env` change, or MySQL not started.	`php artisan config:clear`; start MySQL.

Everything slow with many children	One scoring lane and a CPU-only Whisper `small`. This is expected.	Stagger groups of 5–8; tell teachers the board's estimate is the real wait.
Laptop sleeps mid-session	Power plan.	Plug in; set sleep to Never; disable the lid-close action.

10. After the offline testing → parents on the VPS

1. Take a final backup of the laptop DB and `storage/app` (section 7, step 4).
2. **Back up the VPS database first.** It currently holds only passages, interventions and similar content.
3. The VPS must use the **same** `APP_KEY` as the laptop: user emails and names are stored *encrypted*, and a different key makes them unreadable.
4. Run the same migrations on the VPS, then import the laptop dump and copy the audio files. Rebuild all three caches afterwards (`config:cache`, `route:cache`, `view:cache`) — skipping `route:cache` after a pull gives a 500 inside a Blade view.
5. Confirm the weakness model version matches and migrate before analysing.
6. Verify one parent login and one results page before handing credentials to parents.

> The laptop is the source of truth for the test week. Push one way only; do not edit records on both sides.

Generated for the Old Sagay Elementary School system test. Paths and the IP above reflect the setup on 2026-10-22.